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| Tool Version : Vivado v.2024.1 (win64) Build 5076996 Wed May 22 18:37:14 MDT 2024
| Date         : Wed Sep 16 23:05:46 2026
| Host         : MICHALADENND5D2 running 64-bit major release  (build 9200)
| Command      : report_utilization -file lab2_top_utilization_synth.rpt -pb lab2_top_utilization_synth.pb
| Design       : lab2_top
| Device       : xc7a100tcsg324-1
| Speed File   : -1
| Design State : Synthesized
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Utilization Design Information

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1. Slice Logic

Site Type	Used	Fixed	Prohibited	Available	Util%
Slice LUTs*	8	0	0	63400	0.01
LUT as Logic	8	0	0	63400	0.01
LUT as Memory	0	0	0	19000	0.00
Slice Registers	0	0	0	126800	0.00
Register as Flip Flop	0	0	0	126800	0.00

Register as Latch	0	0	0	126800	0.00
F7 Muxes	0	0	0	31700	0.00
F8 Muxes	0	0	0	15850	0.00

* Warning! The Final LUT count, after physical optimizations and full implementation, is typically lower. Run opt_design after synthesis, if not already completed, for a more realistic count.
Warning! LUT value is adjusted to account for LUT combining.
Warning! For any ECO changes, please run place_design if there are unplaced instances

1.1 Summary of Registers by Type

Total	Clock Enable	Synchronous	Asynchronous
0	_	-	-
0	_	-	Set
0	_	-	Reset
0	_	Set	-
0	_	Reset	-
0	Yes	-	-
0	Yes	-	Set
0	Yes	-	Reset
0	Yes	Set	-
0	Yes	Reset	-

2. Memory

Site Type	Used	Fixed	Prohibited	Available	Util%
Block RAM Tile	0	0	0	135	0.00
RAMB36/FIFO*	0	0	0	135	0.00
RAMB18	0	0	0	270	0.00

* Note: Each Block RAM Tile only has one FIFO logic available and therefore can accommodate only one FIFO36E1 or one FIFO18E1. However, if a FIFO18E1 occupies a Block RAM Tile, that tile can still accommodate a RAMB18E1

3. DSP

Site Type	Used	Fixed	Prohibited	Available	Util%
DSPs	0	0	0	240	0.00

4. IO and GT Specific

Site Type	Used	Fixed	Prohibited	Available	Util%
Bonded IOB	51	0	0	210	24.29
Bonded IPADs	0	0	0	2	0.00
PHY_CONTROL	0	0	0	6	0.00
PHASER_REF	0	0	0	6	0.00
OUT_FIFO	0	0	0	24	0.00
IN_FIFO	0	0	0	24	0.00
IDELAYCTRL	0	0	0	6	0.00
IBUFDS	0	0	0	202	0.00
PHASER_OUT/PHASER_OUT_PHY	0	0	0	24	0.00
PHASER_IN/PHASER_IN_PHY	0	0	0	24	0.00
IDELAYE2/IDELAYE2_FINEDELAY	0	0	0	300	0.00
ILOGIC	0	0	0	210	0.00
OLOGIC	0	0	0	210	0.00

5. Clocking

Site Type	Used	Fixed	Prohibited	Available	Util%
BUFGCTRL	0	0	0	32	0.00
BUFIO	0	0	0	24	0.00

MMCME2_ADV	0	0	0	6	0.00
PLLE2_ADV	0	0	0	6	0.00
BUFMRCE	0	0	0	12	0.00
BUFHCE	0	0	0	96	0.00
BUFR	0	0	0	24	0.00
+------------+------+-------+------------+-----------+-------+

6. Specific Feature

+-------------+------+-------+------------+-----------+-------+
| Site Type | Used | Fixed | Prohibited | Available | Util% |
+-------------+------+-------+------------+-----------+-------+
BSCANE2	0	0	0	4	0.00
CAPTUREE2	0	0	0	1	0.00
DNA_PORT	0	0	0	1	0.00
EFUSE_USR	0	0	0	1	0.00
FRAME_ECCE2	0	0	0	1	0.00
ICAPE2	0	0	0	2	0.00
PCIE_2_1	0	0	0	1	0.00
STARTUPE2	0	0	0	1	0.00
XADC	0	0	0	1	0.00
+-------------+------+-------+------------+-----------+-------+

7. Primitives

+----------+------+---------------------+
| Ref Name | Used | Functional Category |
+----------+------+---------------------+
OBUF	32	IO
IBUF	19	IO
LUT4	8	LUT
LUT5	7	LUT
+----------+------+---------------------+

8. Black Boxes

+----------+------+
| Ref Name | Used |
+----------+------+

9. Instantiated Netlists

+----------+------+
| Ref Name | Used |
+----------+------+